SEVEN WAVES OF MOBILE COMPUTING

The history of mobile computing can be divided into a number of eras, or waves, each characterized by a particular technological focus, interaction design trends, and by leading to fundamental changes in the design and use of mobile devices.

The history of mobile computing can be understood through **seven waves of development**, where each wave represents a major shift in technology, design, and usage of mobile devices. These waves are not strictly sequential, but they show how mobile computing has evolved into modern smart systems that we use today.

1. Portability
2. Miniaturization
3. Connectivity
4. Convergence
5. Divergence
6. Apps
7. Digital ecosystems

Portability

Portability refers to the ability of computing devices to be easily carried and used anywhere. Early mobile devices were designed to move away from fixed desktop systems and support communication on the go. This wave introduced the basic idea of mobility in computing. It laid the foundation for all future mobile technologies.

The era of focus on *Portability* was about reducing the size of hardware to enable the creation of computers that could be physically moved around relatively easily.

Miniaturization

*M*iniaturization is the process of making devices smaller while increasing their performance and efficiency. Advances in hardware technology allowed computers and phones to shrink in size without losing functionality. This made it possible to carry powerful devices in the pocket. It was a key step toward modern smartphones.

Miniaturization was about creating new and significantly smaller mobile form factors that allowed the use of personal mobile devices while on the move.

By the early 1990s, the size of computer hardware had reached a point that allowed radically new and smaller form factors of mobile computers to evolve and emerge on the market. These predominantly handheld devices were labelled palmtop computers, digital organizers, or "Personal Digital Assistants" (PDAs). PDAs differed from laptop PCs by being truly mobile and something that the users could operate while actually moving around physically. They were not thought of as alternatives to desktop or laptop computers, but rather as small and lightweight supplemental devices for busy businessmen spending some of their time away from their PC. Whereas the focus of laptop computing was predominantly on portability and mobile access to documents and applications available on desktop computers, palmtop computing introduced an additional focus on applications and interaction styles designed specifically for mobile devices and mobile users.

Connectivity

Connectivity focuses on enabling devices to communicate through wireless networks such as cellular networks, Wi-Fi, and Bluetooth. This wave allowed mobile devices to access the internet and share information in real time. It transformed mobile phones from standalone devices into connected systems. Communication became faster and more global.

Connectivity was about developing devices and applications that allowed users to be online and communicate via wireless data networks while on the move.

With the introduction of the digital Global System for Mobile Communications (GSM) mobile phone system in 1991, which also included the Short Message Service (SMS) communication component, the complexity and functionality of handsets began evolving rapidly. So did the uptake of mobile phone technology by the broad population worldwide. This meant that mobile phone developers were suddenly faced with a huge challenge of interaction design not only for making phone calls, but also for handling contacts, calendars, text-based messages, and browsing the Internet.

Convergence

Convergence refers to combining multiple technologies and functions into a single mobile device. Phones began to integrate cameras, music players, GPS, and internet browsing features. This reduced the need for separate devices for different tasks. It made smartphones powerful all-in-one tools.

Convergence was about integrating emerging types of digital mobile devices, such as Personal Digital Assistants (PDAs), mobile phones, music players, cameras, games, etc., into hybrid devices.

One of the most interesting eras of mobile computing began when different types of specialized mobile devices began converging into new types of *hybrid* devices with fundamentally different form factors and interaction designs. The first phase of this was the emergence of "smart phones", which combined the functionality of a PDA with that of a mobile phone. The development of smart phones involved exploration of a wide range of form factors and interaction designs and led to a series of innovative solutions. Many of these involved designs where the physical shape of the device could be changed depending on what the user wanted to use it for. Other designs, like the Blackberry, introduced a "wide-body mobile phone" form factor with a PDA size display and a miniature QWERTY keyboard in place of the traditional 12-key numeric keypad.

The second phase of convergence combined mobile phones with various rich media capabilities, such as digital cameras, music players, video recording and playback, and television and radio reception. Whereas smart phones were attractive for business professionals' work activities and productivity, multimedia phones were attractive for everyday people's leisure, fun and socializing.

Divergence

Divergence describes the rise of specialized mobile devices and platforms for different user needs. Instead of one universal device, mobile technology started to branch into wearables, tablets, and smart devices. Each device is optimized for specific functions and user experiences. This wave increased flexibility in mobile computing.

Divergence took an opposite approach to interaction design by promoting information appliances with specialized functionality rather than generalized ones.

Contrasting the convergence approach, the trend of *divergence* suggested a single function/many device or "information appliance" approach where each device is "designed to perform a specific activity, such as music, photography, or writing". The driving force behind this line of thought is that having a wide range of good specialized tools is better than a general one that does not perform any task particularly well. Specialized tools facilitate optimization of functionality over time and refinement of well-known paradigms of use. The fundamental view promoted by the trend of divergence is that mobile user experience is *inversely* proportionate to the functional scope of interactive mobile devices and systems: "less is more".

Apps

The apps wave introduced mobile applications that run on operating systems like Android and iOS. Users could now download and install software for specific tasks such as banking, gaming, and communication. This shifted mobile devices into customizable platforms. Apps became the core of mobile user experience.

The latest wave of *apps* is about developing matter and substance for use and consumption on mobile devices, and making access to this fun or functional interactive application content easy and enjoyable.

In June 2007, Apple launched the iPhone. Like many of its contemporaries this was a converged mobile device functioning as a camera phone, portable media player, and Internet client with e-mail, web browsing, and high-speed wireless network connectivity. However, rather than being just another incremental step in the evolution of converged mobile devices, the iPhone represented a significant rethinking of the design of mobile interactions and a series of notable interaction design choices. It featured a large high-resolution capacitive multi-touch display with simple gesture capabilities, such as swiping and pinching, and departed completely from the predominant use of physical keys and a stylus for text entry and interaction. Instead of navigating large and deep hierarchies of menus, the user experience was much more fluid and aesthetic, and the phone was both extremely easy and pleasurable to use. The iPhone also featured a number of embedded context sensors, which changed the orientation mode of the display depending on how

it was held and it thereby changed the mode of the phone application when held close to the face during a call. The later inclusion of GPS and a digital compass extended this "context-awareness" capability to also

Digital ecosystems

Digital ecosystems refer to interconnected systems of devices, services, and platforms working together through cloud computing and the internet. Mobile devices now connect with smart homes, wearables, and online services seamlessly. Data flows across multiple platforms to provide personalized experiences. This wave represents the modern era of intelligent, connected computing.

Finally, the emerging wave of *digital ecosystems* is about the larger wholes of pervasive and interrelated technologies that interactive mobile systems are increasingly becoming a part of.

As we move into the second decade of the new millennium, the challenges facing mobile computing and interaction design continue to evolve. The technical capabilities of our mobile devices have improved significantly to the point where factors such as screen, input capabilities, processing power, network speed, and battery lifetime are much less of an issue than only half a decade ago. At the same time, we have also become sufficiently skilled at designing for relatively small screens and for the different input capabilities of mobile devices so that millions of ordinary people are actually able to download and use the applications being developed, and are even willing to pay for some of them. To a large extent, therefore, we have now successfully solved the majority of problems facing mobile interaction researchers and designers in the past. However, as the history of all areas of computing have shown us, it is highly unlikely that we have reached an end point. As in the past, the technology and interaction design we are witnessing today is just the starting point for the continuing evolution of the technology and interaction design of tomorrow. But what are then the challenges and opportunities for the design of mobile interactions to come? What will the next wave of mobile computing be about?

Looking on the current trends, it appears that the next wave of mobile computing and interaction design is going to be about the creation of *digital ecosystems* in which mobile computing plays a central role in concert with other ubiquitous computing resources. This challenges us to move beyond considering interactive mobile devices, systems, and services as entities that can

meaningfully be designed and studied in isolation from the larger use context or artefact ecologies that they are a part of.

Example: Smartphone

• Portability: carry it anywhere you want

• Miniaturization: make it possible to build device to fit in your pocket

• Connectivity: Wi-Fi, LTE/4G, cellular, Bluetooth

• Convergence: phone, camera, gaming device, movie streaming, music player, …

• Divergence:

• Applications: “Rise of the Apps”

• Digital Ecosystem: social networks, distributed gaming, video streaming, work apps, …